

SAMY RJOB

Software Engineer · Java, Spring Boot & AI Systems

Manchester, UK · samy_rjob@proton.me · +33 6 52 05 56 09 · samyjob.is-a.dev · github.com/samyjob

PERSONAL STATEMENT

Software engineer with an MSc in Computer Science, building software since 2022. My main backend is Java 21 and Spring Boot 3: I designed and built a Spring Boot platform of around 92,000 lines, including an AI agent runtime with LLM tool calling, MCP connectors and human-in-the-loop approvals. I work where backend engineering meets applied AI, with security designed in from the start.

TECHNICAL SKILLS

Java & Spring: Java 21, Spring Boot 3, Spring Security (JWT), Spring Data JPA / JDBC, Spring Kafka, WebSocket, GraphQL, Flyway, Maven, JUnit 5, Mockito

AI & LLM: LLM tool calling and agent runtimes, MCP, RAG, embeddings, vector databases (Qdrant), Ollama, OpenAI API

Data & Infra: MySQL, PostgreSQL, Redis, Kafka, Docker, AWS (S3 / SES / CloudFront / KMS), CI/CD (GitHub Actions)

Also: TypeScript, Python, Go, SQL · Angular, React · JWT, WebAuthn, HMAC, Ed25519, Row-Level Security

PROJECTS

Futura · Java & Spring Boot Platform with AI Agents · futura-game.com

Jan 2025 – Present

A multiplayer world with an integrated CRM whose AI agents appear in-world as characters. Built and run solo.

- Built the Java 21 / Spring Boot 3.4 backend (~92,000 lines): REST, WebSocket and GraphQL APIs, Spring Security with JWT, Spring Data JPA with Flyway migrations, Redis caching, Spring Kafka and AWS S3 / SES, covered by 77 JUnit 5 and Mockito test classes.
- Built an AI agent runtime in Java: an LLM tool-calling loop, a custom agent-definition language with a hand-written parser and compiler, MCP connectors to external tools, and human-in-the-loop approvals so an owner signs off before an agent acts.
- Event-driven, modular design: a Kafka audit trail of in-game item trades, decoupled from trade execution, and a hexagonal (ports-and-adapters) domain isolated from its database and LLM adapters.

2YM Chatbot · Production RAG Assistant (MSc project)

Nov 2025 – Present

A RAG assistant for university students and programme leaders, used by real cohorts.

- Answers only above a 0.65 confidence threshold and escalates to a human below it, so it never guesses: a 9.4-second median answer versus a ~2-day staff wait, with 67.9% of conversations resolved without staff.

EXPERIENCE

Elite Talents · Freelance Backend Developer

2025 – May 2026

Toulouse, France (Remote) · elitetalents.fr

- Designed and built the whole backend of a live recruitment platform: a 165-endpoint Next.js 14 API in TypeScript with Zod-validated inputs, spanning six actor types on a 50-table PostgreSQL schema with row-level security on every table (default-deny).
- Own the security model and integrations: a two-tier Supabase access model (service-role client kept strictly server-side), WebAuthn passkeys, JWT, bcrypt, CSRF protection and rate limiting, plus Stripe billing via the Supabase Wrapper (API key in Vault) and OpenAI candidate-to-role matching.

YES Germany · Software Developer Intern

Apr 2024 – Sep 2024

Faridabad, India

- Launched an AI-powered generator for statements of purpose (SOPs) using the OpenAI API, cutting creation time from 4 hours to 15 minutes per SOP; deployed on Streamlit.
- Built an ETL pipeline extracting 3,500+ SOPs via the Dropbox API with spaCy classification (85% accuracy), and automated Dropbox / Google Sheets workflows that cut the admissions team's manual work by 70%; mentored a trainee.

EDUCATION

MSc Computer Science · University of Chester, Chester, UK

2024 – 2026

Master, Programme Grande École · Audencia Business School, Nantes, France

2020 – 2024

CPGE (Scientific & Commercial Preparatory Classes) · Lycée Malherbe, Caen, France

2018 – 2020